MSc Data Science @ UNIMIB

June 2026

Europe's Aging Population

A demographic crisis
reshaping the continent

Data source: Eurostat

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Research questions

Europe's demographic shift is one of the most profound and slow moving crises of our time. To navigate its complexity, this analysis is structured around three core research questions, each addressing a different dimension of the phenomenon: its scale, its causes, and its possible responses.

01.

How old is Europe really getting?

This first question explores the scale and geography of demographic aging across EU member states, tracing how the age structure of European populations has evolved over the past six decades and whether the trend is shared equally across the continent.

02.

What are the structural causes?

Two converging forces drive demographic aging: fewer births and longer lives. Their interaction is reshaping Europe's age structure, with profound consequences for pension systems and public finances.

03.

Can migration reverse the trend?

As natural population growth weakens across much of the continent, migration has emerged as a critical demographic variable. Yet its contribution varies enormously by country, and raises a fundamental question: is migration enough to sustain Europe's population in the long run?

Data overview

All data used in this analysis comes from **Eurostat**, the official statistical office of the European Union. Eurostat provides high-quality, harmonized demographic data collected using standardized methodologies across all member states, ensuring full comparability between countries and over time.The datasets were downloaded in May 2026 directly from the Eurostat Data Browser portal. All data is available under **Eurostat's open data license**, which allows for **free reuse with attribution** of the source. The decision to rely exclusively on Eurostat data was driven by the **limitations encountered in alternative sources**: many other demographic datasets presented **missing values**, **inconsistent structures**, or **non-harmonized collection methodologies** across countries, making cross-national comparisons unreliable. Eurostat, by contrast, offers a fully standardized and comparable framework across all EU member states. Overall, the data quality was high and no major inconsistencies were detected across the datasets. However, two limitations are worth noting. First, not all EU member states provided complete data across every demographic indicator. To ensure consistency and full comparability across all analyses, the decision was made to **focus exclusively on EU27 member states**, as data coverage for this group was complete and consistent across all datasets used. Second, in some cases the **most recent available year varied** across countries: where a country's data ended earlier than others, the **most recent available observation** was used rather than excluding the country entirely from the analysis. This inconsistency is partly attributable to **reporting lags** inherent in large-scale statistical data collection: some member states publish their national statistics with a delay, meaning that the most recent years may be **incomplete or subject to future revision**. These two issues represent the main data quality challenges encountered, and the methodological choices made to address them are reflected throughout the presentation.

Dataset:

1. **demo_pjanind**, median age & dependency ratio
2. **demo_pjangroup**, population by age group
3. **demo_find**, total fertility rate
4. **demo_mlexpec**, life expectancy at birth
5. **spr_exp_pens**, pension expenditure % GDP
6. **demo_gind**, natural growth & net migration

Methodology

The data was downloaded in TSV format from the Eurostat portal and processed entirely in Python, using pandas and numpy for data cleaning, restructuring, and exploratory analysis

Before proceeding with the analysis, each dataset underwent a preprocessing phase including filtering of relevant time ranges and EU27 member states, reshaping from wide to long format to enable time-series analysis, and mapping of Eurostat country codes to full country names for visualization purposes. Claude (Anthropic) was also used as an AI assistant to support parts of this preprocessing workflow, in particular for the conversion of country codes into full names compatible with the visualization tools.

The analysis was conducted on two complementary levels: the first analysis compared countries in the most recent year available, while the second analysis examined the evolution of key demographic indicators from 1960 to the present.

The visualizations were produced using Python (Matplotlib, Seaborn, Geopandas) and Datawrapper, a professional tool used by international news outlets such as the NYT and The Economist. The choice between the two tools was guided by the nature of each chart and the communicative effectiveness of the final result.

To ensure full reproducibility, all code, charts, and datasets have been made publicly available on GitHub at: https://github.com/AndreaPorro/Data-visualization

Insights from the Data

01.

demo_pjanind: Using descriptive statistics and ranking, Italy emerged as the oldest EU27 country (median age: 49.1 years), while Ireland is the youngest (39.6). No country falls below 39, confirming aging is continent-wide. The EU27 average rose from 31.5 years in 1960 to 43.8 in 2025, computed via grouped time-series averaging.

demo_pjangroup: Through cross-year comparison, the population pyramid revealed a structural shift: the 80-84 age group nearly doubled (7.3M → 13.7M) between 2001 and 2023, while the 0-4 group shrank from 22.1M to 20.4M.

02.

demo_find: Sorting and threshold comparison revealed that not a single EU27 country reaches the replacement threshold of 2.1. EU27 average stands at 1.45. Europe has been below this threshold since 1978, for 47 consecutive years.

demo_mlexpec + **demo_find**: Merging and correlation analysis produced the demographic scissors chart: life expectancy rose from 68.8 to 80.9 years while fertility fell from 2.60 to 1.45. The two curves cross around 1985–1990.

spr_exp_pens: Ranking and outlier detection showed France spending 12.07% of GDP on pensions, over three times more than Ireland (3.40%), with the EU27 average at 7.9%.

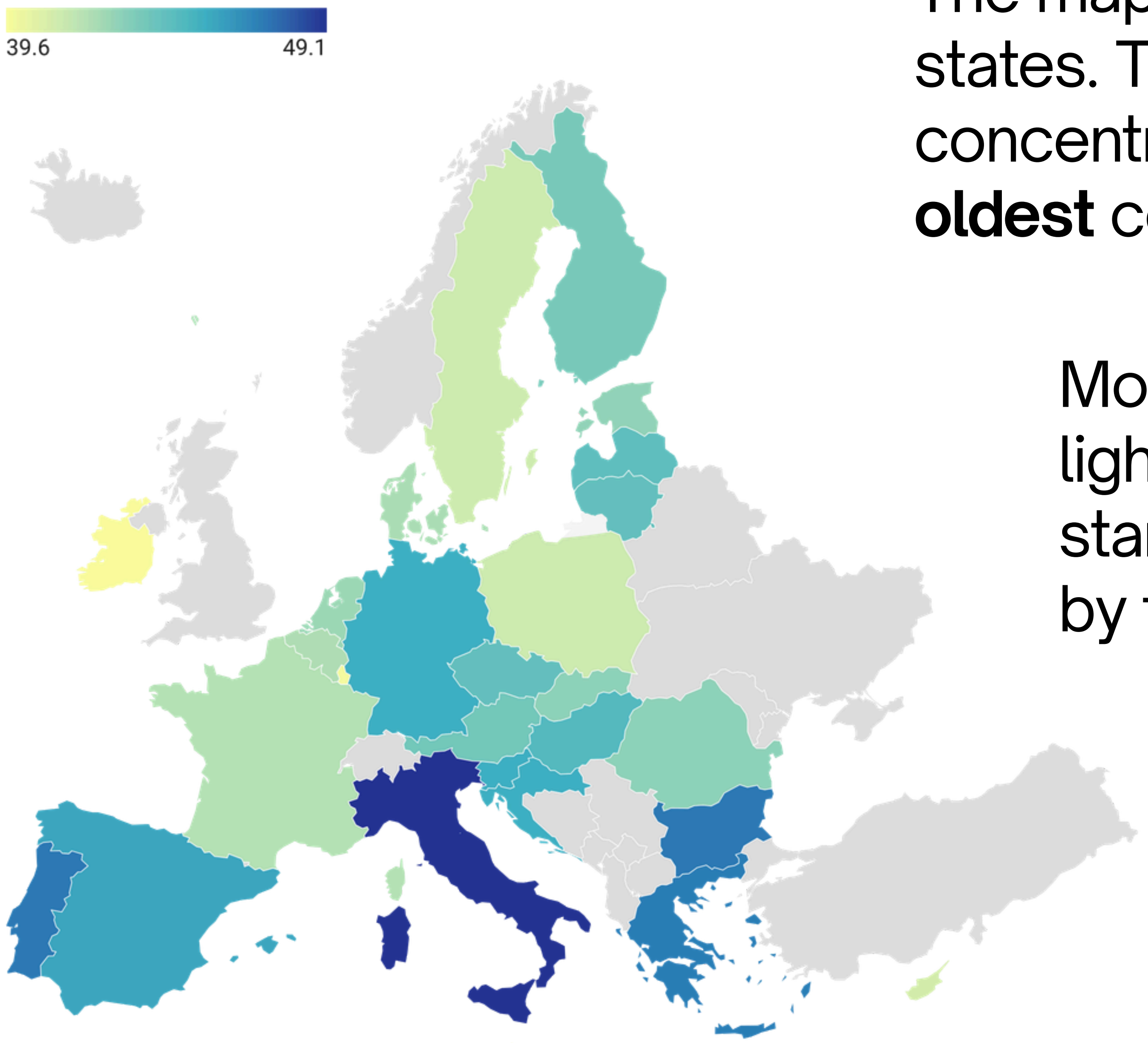
03.

demo_gind: Comparative analysis of natural growth rate vs net migration rate revealed that 20 out of 27 EU countries record negative natural growth. Migration has become the primary demographic driver, with the two rates converging around 1995. Two outlier events are clearly visible: the COVID-19 spike in deaths (2020) and the Ukraine war migration surge (2022).

How old is Europe?
Median age across EU27

Median age by EU27 country (2025)

Southern and Eastern Europe are the regions with the oldest populations



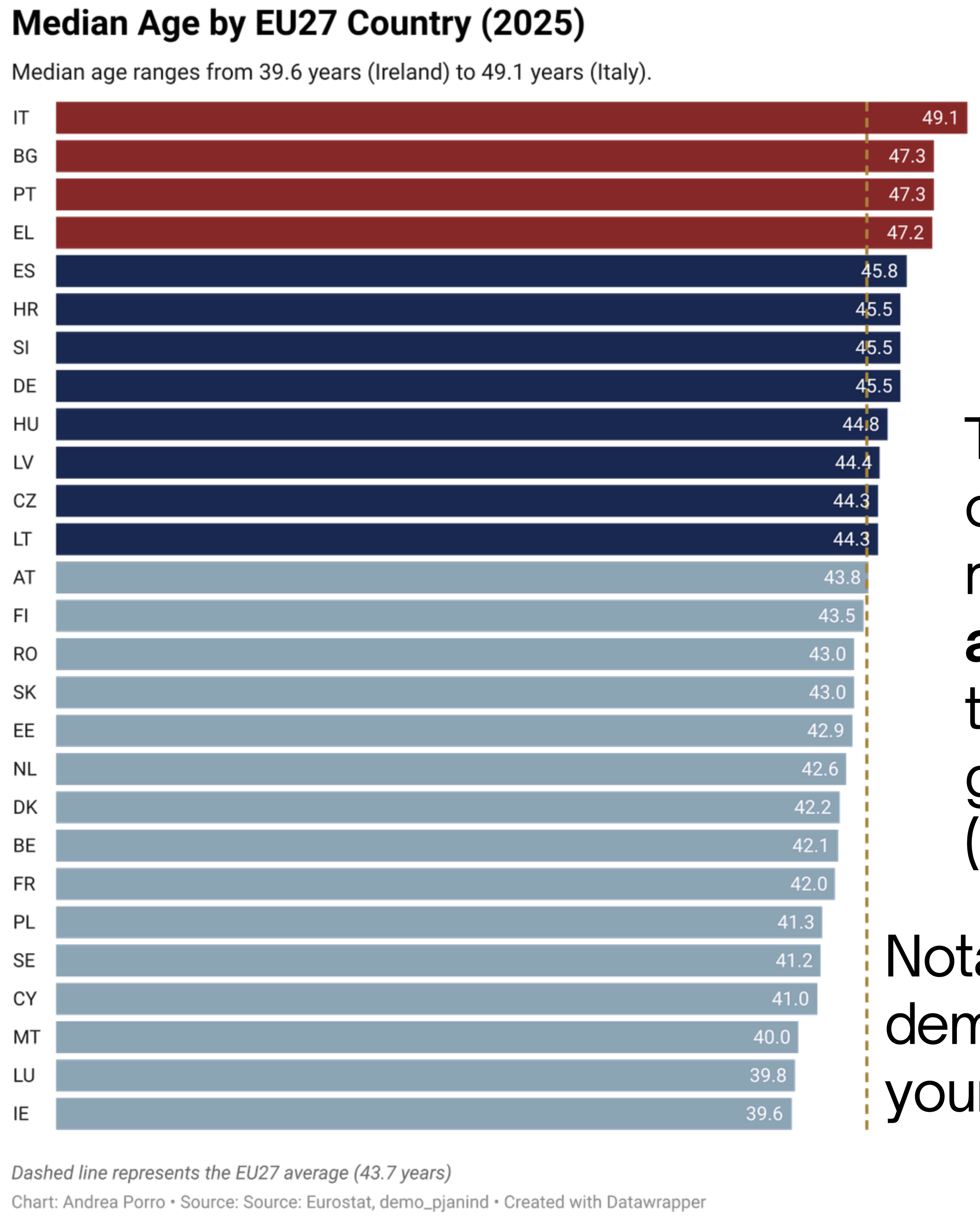
Map: Andrea Porro • Source: Eurostat, demo_pjanind • Created with Datawrapper

The map reveals a clear geographical pattern across EU27 member states. The darkest shades, indicating the highest median ages, are concentrated in **Southern Europe**, with Italy standing out as the **oldest** country on the continent, and in parts of **Eastern Europe**.

Moving **northward** and **westward**, the colours progressively lighten, suggesting **younger** population structures. Ireland stands out visually as the youngest EU27 country, represented by the lightest shade on the map.

Overall, the map suggests that demographic aging is not a uniform phenomenon across Europe, its intensity varies significantly depending on the region, revealing a **continent** that is **aging at** very **different speeds**.

How old is Europe?
Median age across EU27 countries

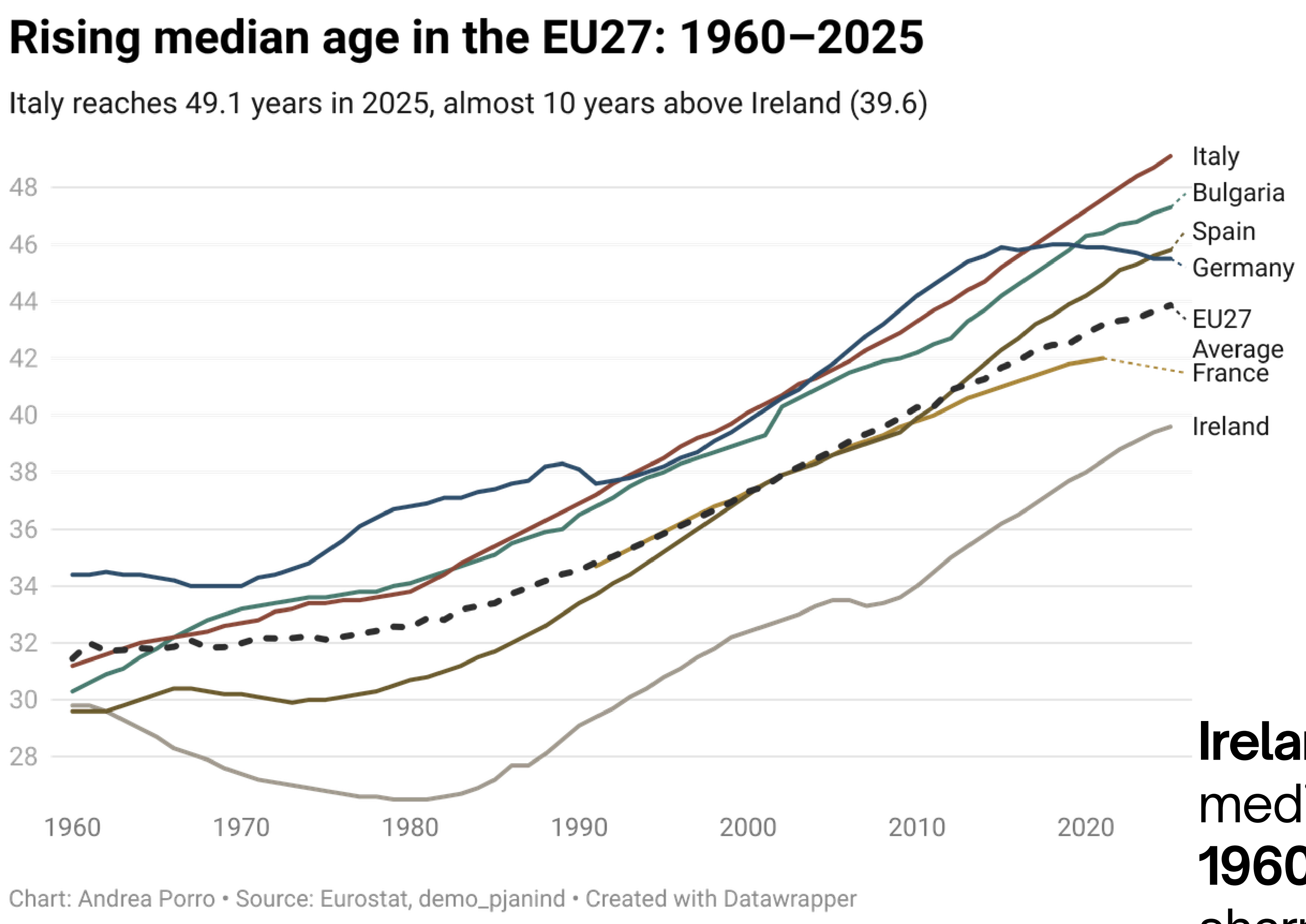


The bar chart confirms and quantifies the geographical pattern observed in the map. With a median age of **49.1 years**, **Italy** stands as the **oldest country** in the EU27, followed closely by **Bulgaria** (47.3), **Portugal** (47.3) and **Greece** (47.2).

Three distinct groups emerge from the data. A first group of countries, concentrated in Southern and Eastern Europe, records median ages above 45 years, well above the **EU27 average of 43.7 years**. A second, larger group clusters around the average, ranging roughly between 42 and 45 years. A third group, including **Ireland** (39.6), **Luxembourg** (39.8) and **Malta** (40.0), maintains significantly **younger population structures**.

Notably, **no EU27 country falls below 39 years**, confirming that demographic aging is a continent-wide phenomenon, even the youngest member states are far older than the global average

How old is Europe?
How Europe has aged: 1960–2025



The chart shows that demographic aging is not a recent phenomenon, it has been a **continuous** and **uninterrupted trend** across all selected countries since the 1960s.

However, the trajectories diverge significantly. Italy has followed the steepest upward path, rising from around **31 years in 1960 to 49.1 years in 2025**, an increase of nearly 18 years. Bulgaria and Spain show similarly sharp accelerations, particularly from the 1990s onwards.

Ireland stands out as a notable exception: its median age actually **declined between the 1960s and 1980s**, before reversing and rising sharply

How old is Europe?
Population pyramid: EU27 2001 vs 2023

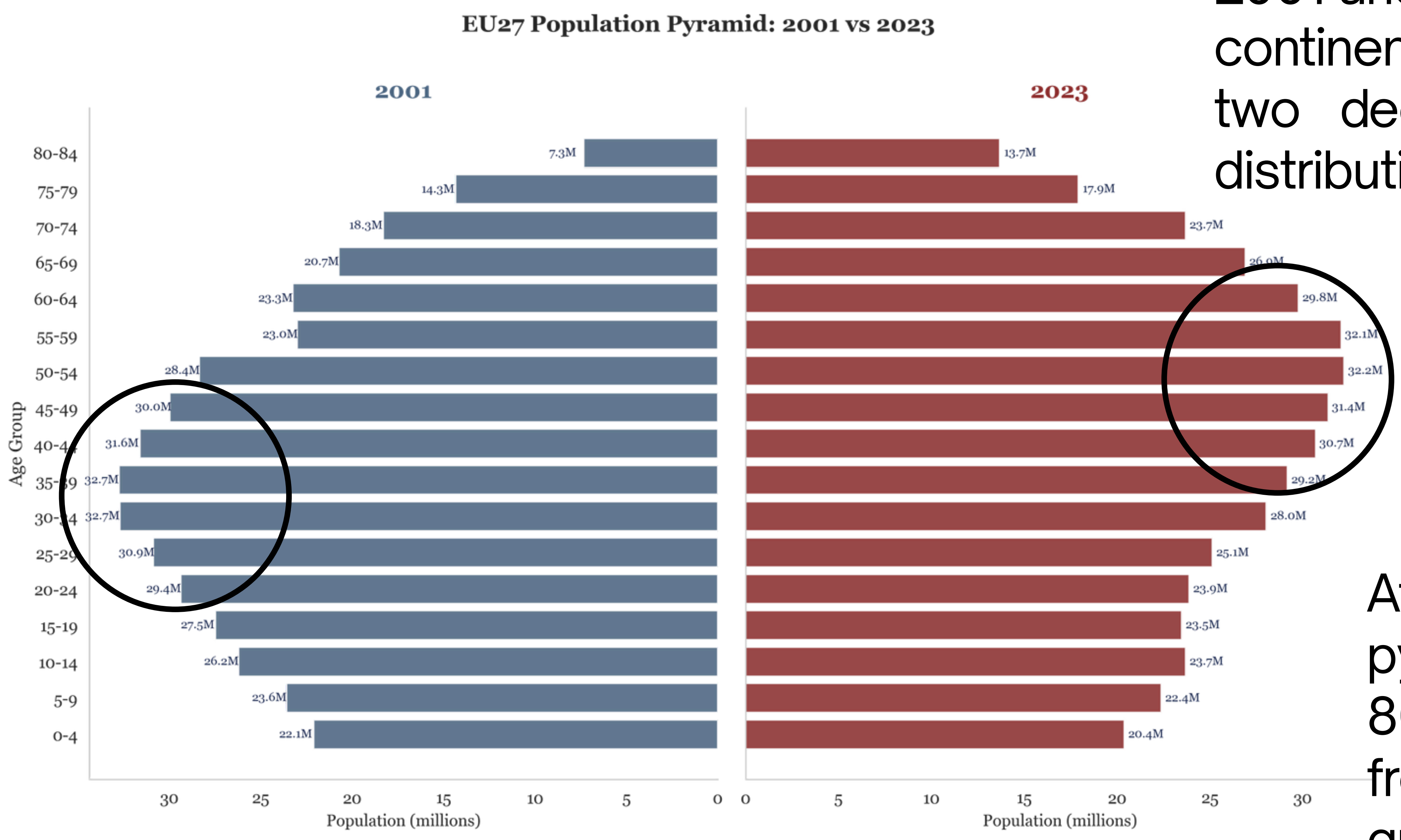


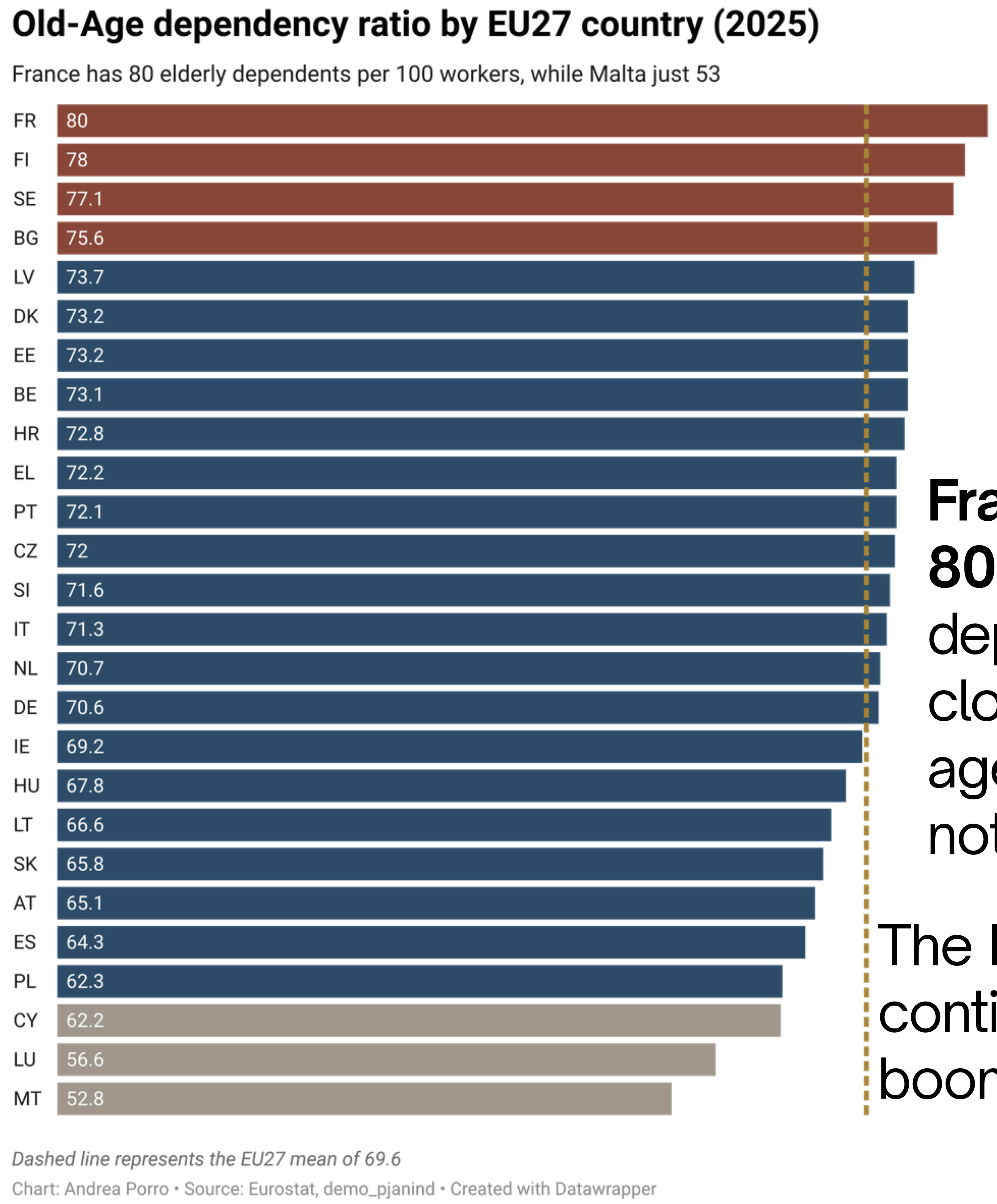
Chart: Andrea Porro • Source: Eurostat • Created with Python/Matplotlib

Comparing the EU27 population structure in 2001 and 2023 reveals a fundamental shift in the continent's demographic balance. In just over two decades, the shape of the population distribution has changed dramatically.

The base of the pyramid, representing the youngest age groups, **has visibly narrowed**. The 0-4 age group reduced from **22.1M to 20.4M**, while the 10-14 group fell from **26.2M to 23.7M.**

At the same time, the upper portion of the pyramid has expanded considerably. The 80-84 age group nearly **doubled**, growing from **7.3M to 13.7M**, while the 65-69 group increased from **20.7M to 26.9M**.

How old is Europe? The dependency ratio

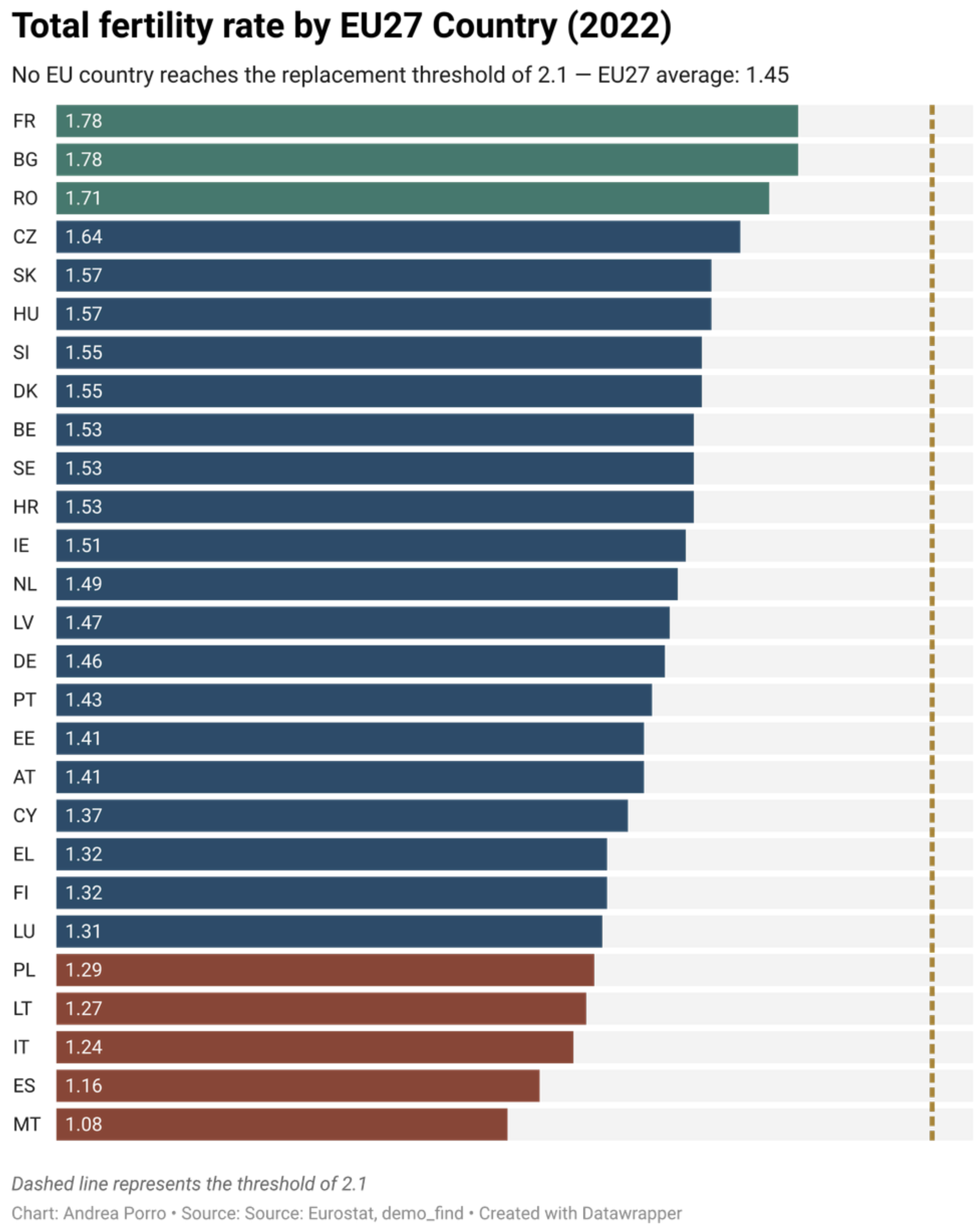


The old-age dependency ratio measures the number of people aged 65 and over per 100 working-age individuals. It translates demographic aging into a concrete economic pressure, the higher the ratio, the greater the burden on the active population to sustain pension and healthcare systems

France records the **highest dependency ratio** in the EU27 at **80.0**, meaning that for every 100 workers, there are 80 elderly dependents to support. **Finland** (78.0) and **Sweden** (77.1) follow closely. Notably, these are not the oldest countries by median age , suggesting that aging pressure on public finances does not always align perfectly with median age rankings.

The **EU27 average** stands at **69.6**, and projections suggest it will continue rising sharply over the coming decades as the baby boom generation reaches retirement age.

What are the structural causes? Fertility rates across EU27 (2022)

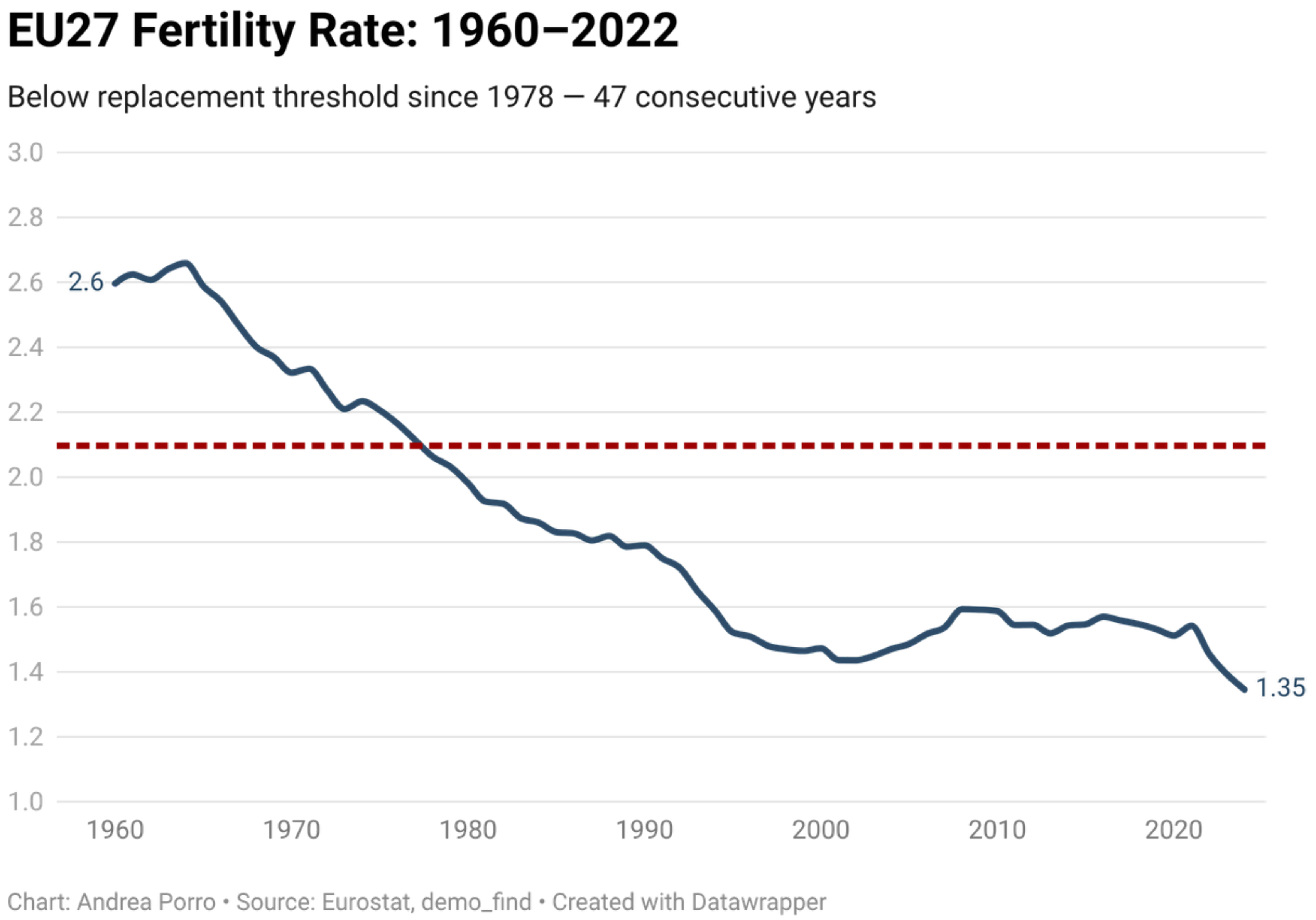


The total fertility rate measures the average number of children a woman is expected to have during her lifetime. A rate of **2.1**, represented by the dashed line, is considered the replacement threshold: the minimum needed to maintain a stable population without migration.

The chart reveals a striking finding: **not a single EU27 country reaches the replacement threshold**. The entire distribution falls to the left of the dashed line, with **the EU27 average** standing at just **1.45**.

Three distinct groups emerge. **France**, **Bulgaria** and **Romania**, represented in green, record the **highest rates**, though still well below 2.1. A large middle group clusters between 1.4 and 1.6. At the **bottom**, **Malta** (1.08), **Spain** (1.16) and **Italy** (1.24), highlighted in terracotta, record the lowest fertility rates in the EU27.

What are the structural causes? The EU27 fertility trend

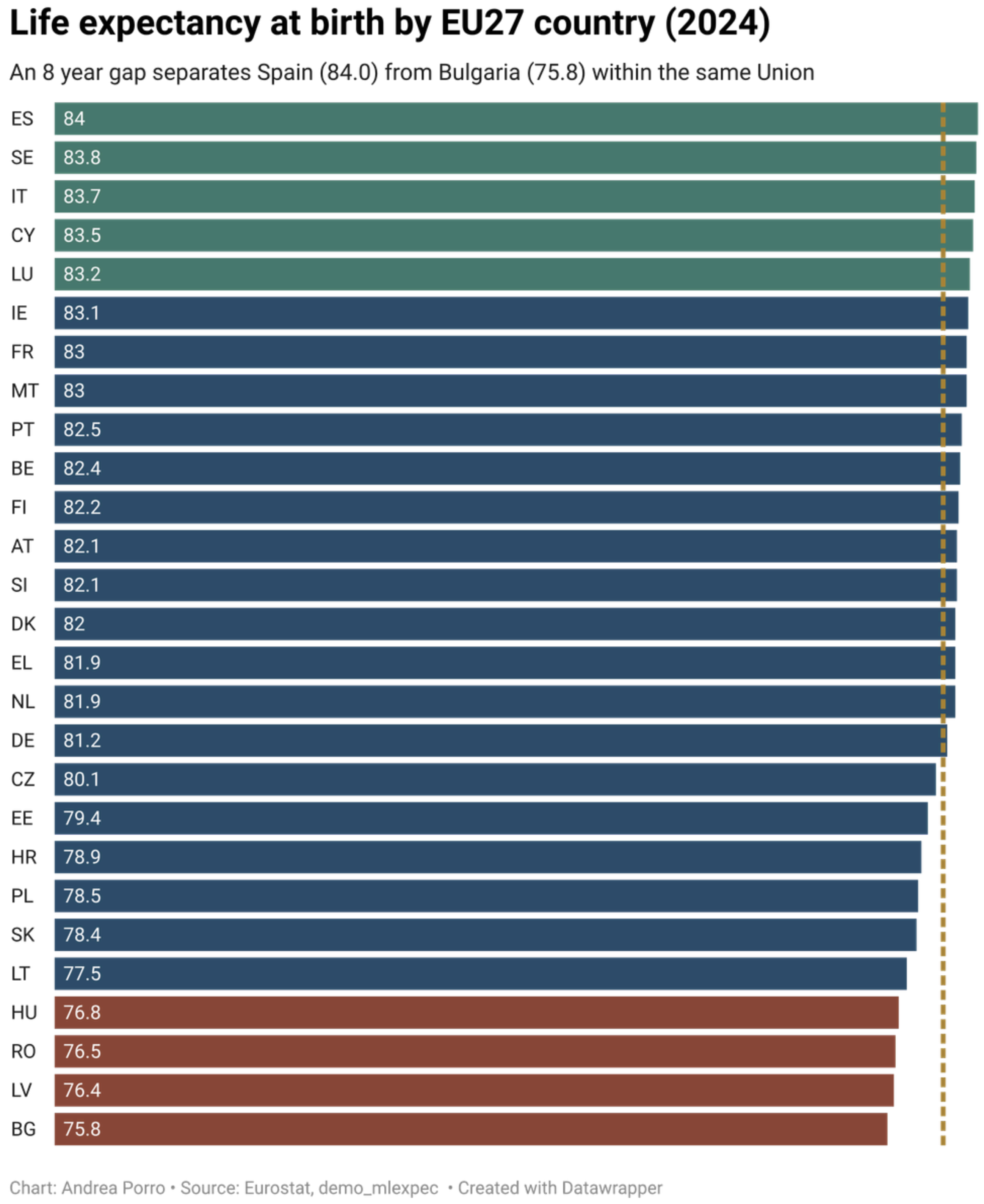


The line chart reveals that the decline in fertility is not a recent phenomenon, it has been a structural and **continuous trend** since the early 1960s. The EU27 average fertility rate fell from **2.60 in 1960** to just **1.45 in 2022**.

The dashed red line marks the **replacement threshold of 2.1**. Europe crossed it for the first time in **1978**, and has never returned above it since. This means that for **47 consecutive years**, the EU27 has not been producing enough births to maintain its population size without external contribution.

The trend is not linear. A modest recovery is visible in the early 2000s, likely reflecting improved economic conditions and expanded family support policies. However, the rate resumed the decline after 2010 and accelerated after the COVID-19 pandemic.

What are the structural causes?
Life expectancy at birth by EU27 (2024)



Life expectancy at birth has reached an **average of 80.9** years across the EU27 in 2024, a remarkable achievement reflecting decades of progress in healthcare, nutrition and living standards.

Three distinct groups emerge from the chart. A first group of five countries, **Spain** (84.0), **Sweden** (83.8), **Italy** (83.7), **Cyprus** (83.5) and **Luxembourg** (83.2), highlighted in green, consistently **leads the ranking**. A large middle group clusters between 78 and 83 years, covering the majority of EU member states. A third group, **Hungary** (76.8), **Romania** (76.5), **Latvia** (76.4) and **Bulgaria** (75.8), highlighted in terracotta, records values significantly **below the EU average**.

Crucially, longer lives contribute directly to demographic aging: as more people survive into old age, the upper portion of the population pyramid expands

What are the structural causes? The demographic scissors

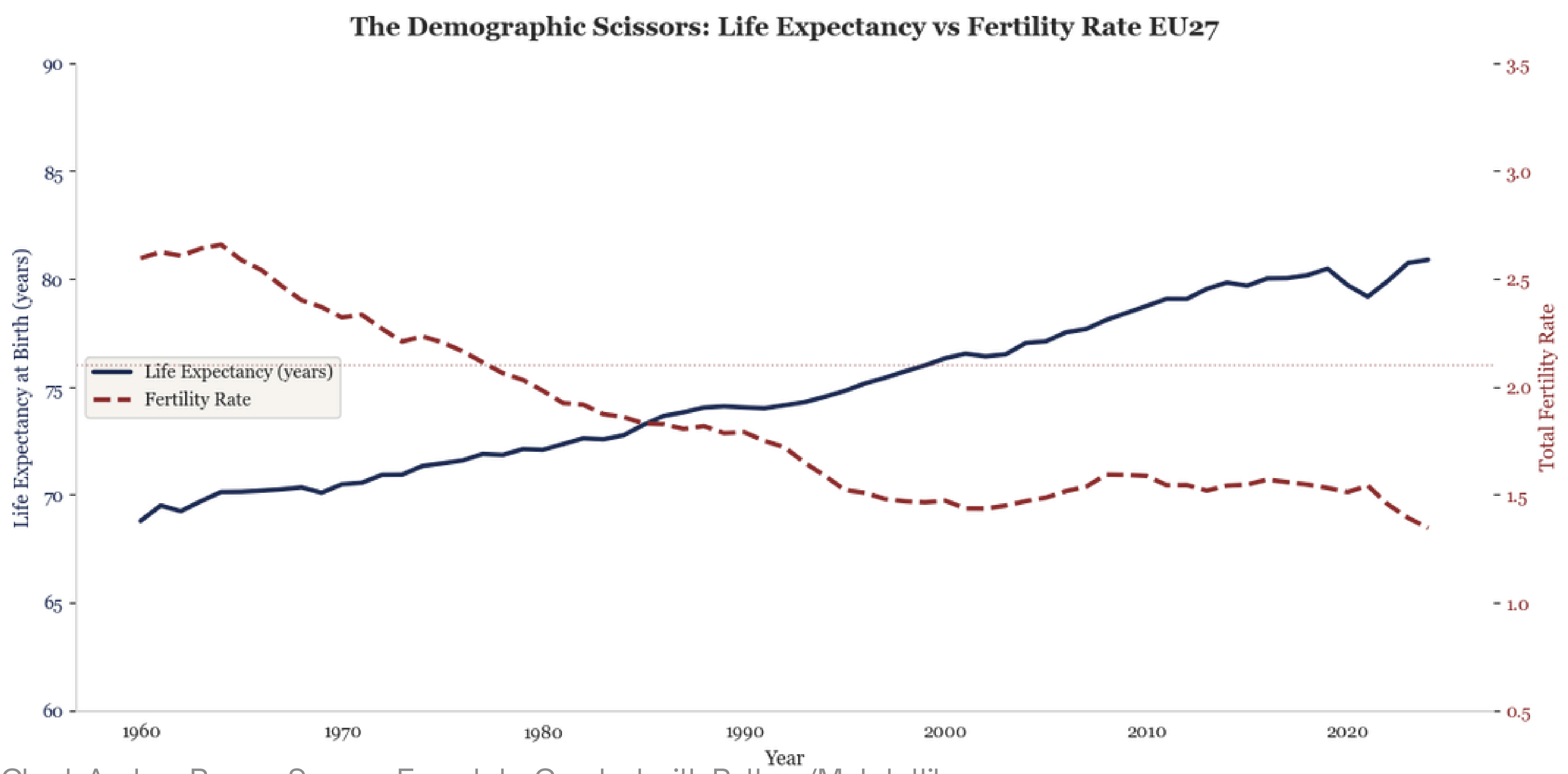


Chart: Andrea Porro • Source: Eurostat • Created with Python/Matplotlib

The chart visualises the two structural forces driving Europe's demographic aging and how they have moved in opposite directions over the past six decades.

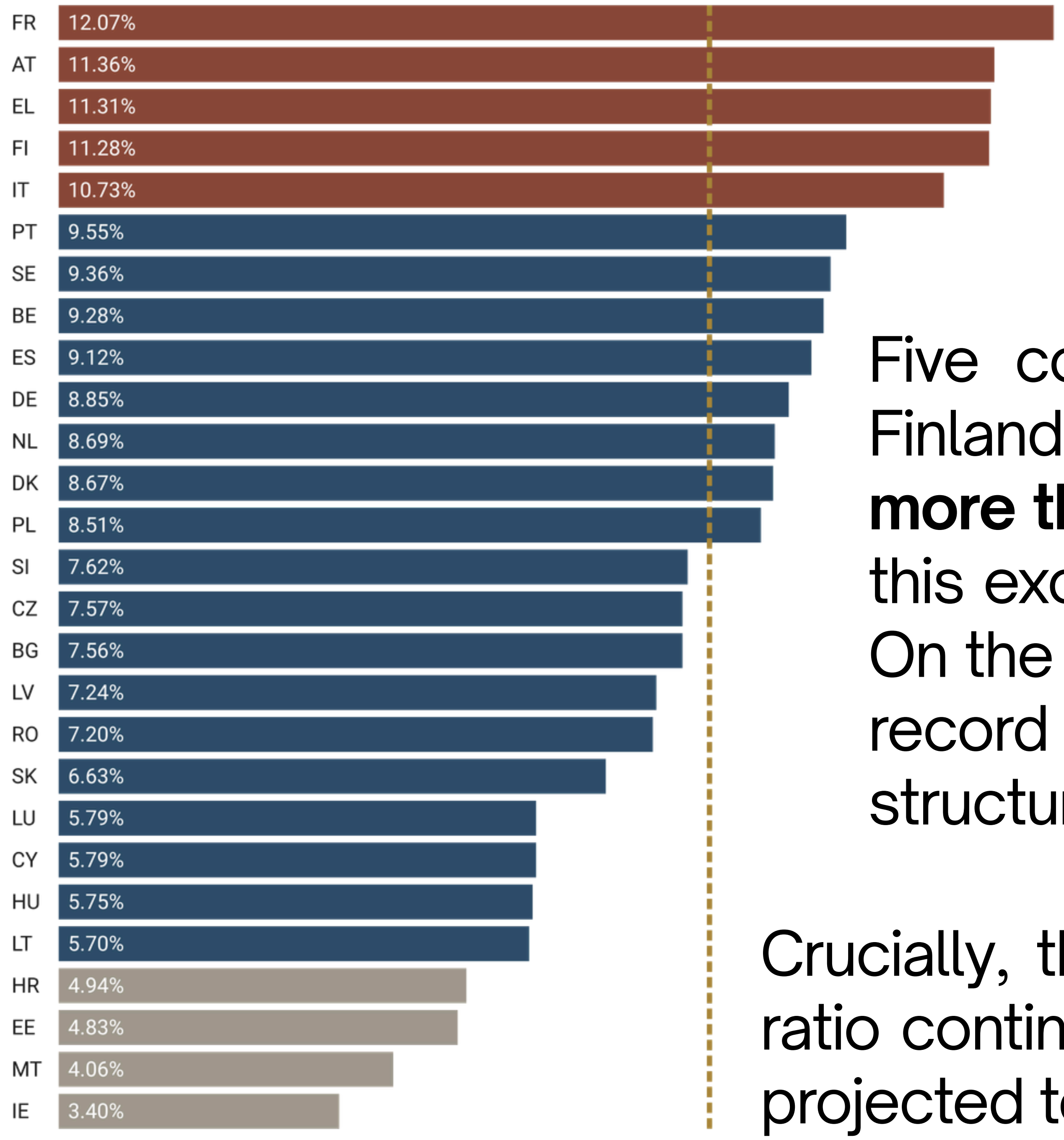
The blue line shows life expectancy at birth, rising steadily **from 68.8 years** in 1960 **to 80.9 years** in 2024. The red dashed line shows the total fertility rate, falling continuously **from 2.60** in 1960 **to** just **1.45** in 2022.

The two curves cross around 1985–1990. From that moment onwards, Europeans have been living significantly longer while having significantly fewer children. Taken together, these two trends create a structural demographic imbalance: a population that lives longer, but does not replace itself. This is the fundamental mechanism behind Europe's aging crisis.

What are the structural causes? The pension burden

Old-Age pension expenditure by EU27 country (2023)

France spends 12.1% of GDP on pensions, while Ireland just 3.4%



Dashed line represents the EU27 mean of 7.9%

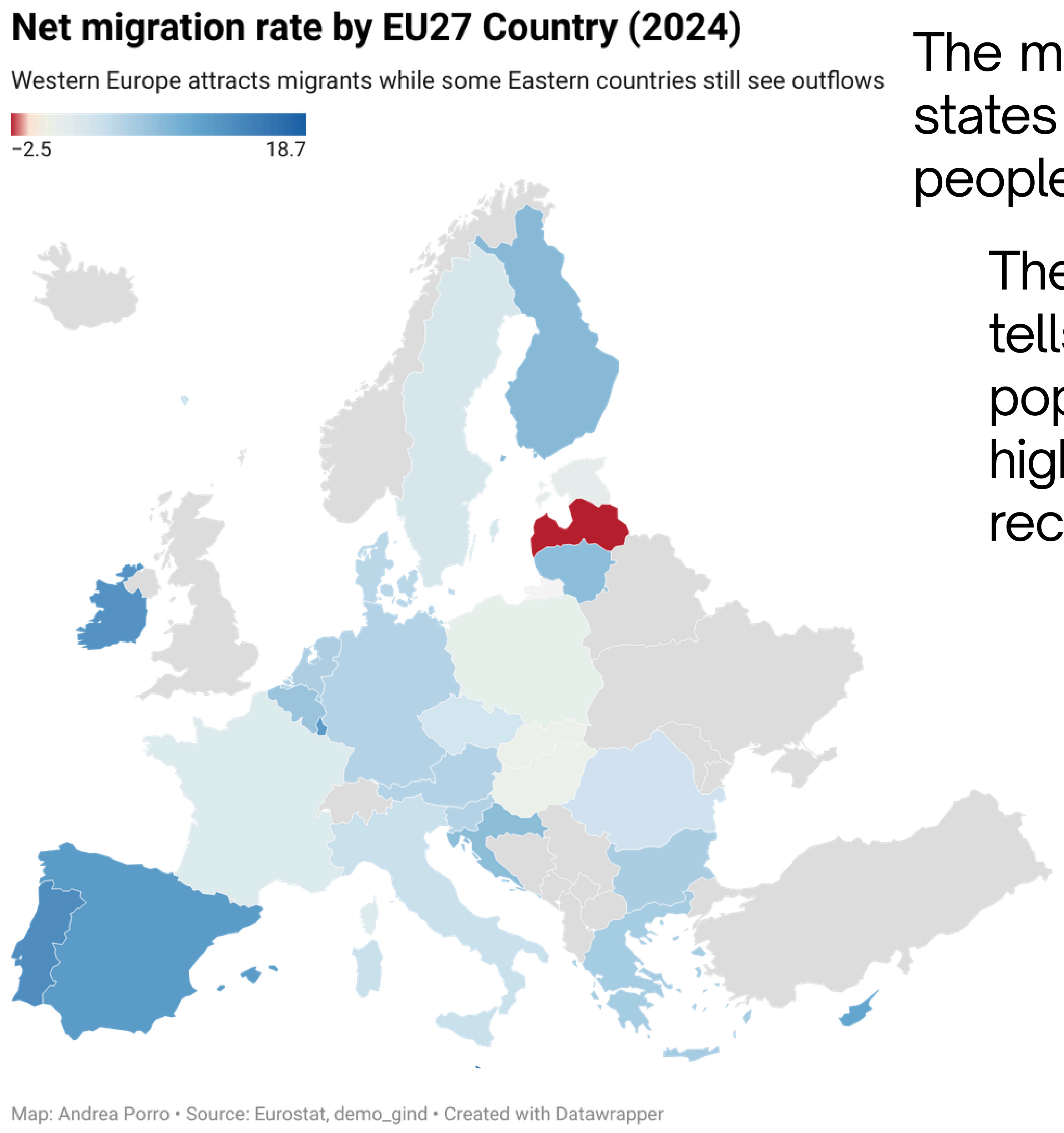
Chart: Andrea Porro • Source: Eurostat, spr_exp_pens • Created with Datawrapper

Pension expenditure as a share of GDP measures the direct fiscal cost of an aging population on national economies. The EU27 average stands at 7.9% of GDP, represented by the dashed line, but the variation across member states is impressive.

Five countries, France (12.07%), Austria (11.36%), Greece (11.31%), Finland (11.28%) and Italy (10.73%), highlighted in terracotta, spend **more than 10% of their GDP** on old-age pensions alone. For context, this exceeds what most countries spend on education and defence. On the other hand, Ireland (3.40%), Malta (4.06%) and Estonia (4.83%) record the lowest pension burdens, reflecting younger population structures and different pension system designs..

Crucially, these figures reflect the situation today. As the dependency ratio continues to rise across all EU27 countries, pension expenditure is projected to increase significantly in the coming decades.

Can migration reverse the trend? The geography of migration



The map shows the net migration rate across EU27 member states in 2024, the difference between people arriving and people leaving, expressed per 1.000 people.

The overwhelming dominance of green across the map tells a clear story: the vast majority of EU27 countries gain population through migration. The darker the shade, the higher the inflow, with Portugal and Spain countries recording the strongest positive rates.

The single exception is Latvia, shown in red, the only EU27 country recording a negative net migration rate in 2024, meaning more people are leaving than arriving.

With the exception of Latvia, migration is flowing positively into every EU27 member state, indicating that, at least directionally, the continent is attracting people rather than losing them.

Can migration reverse the trend? Natural growth vs Net migration by EU27

Natural Growth vs Net Migration by EU27 Country (2024)

20 out of 27 EU countries have negative natural growth

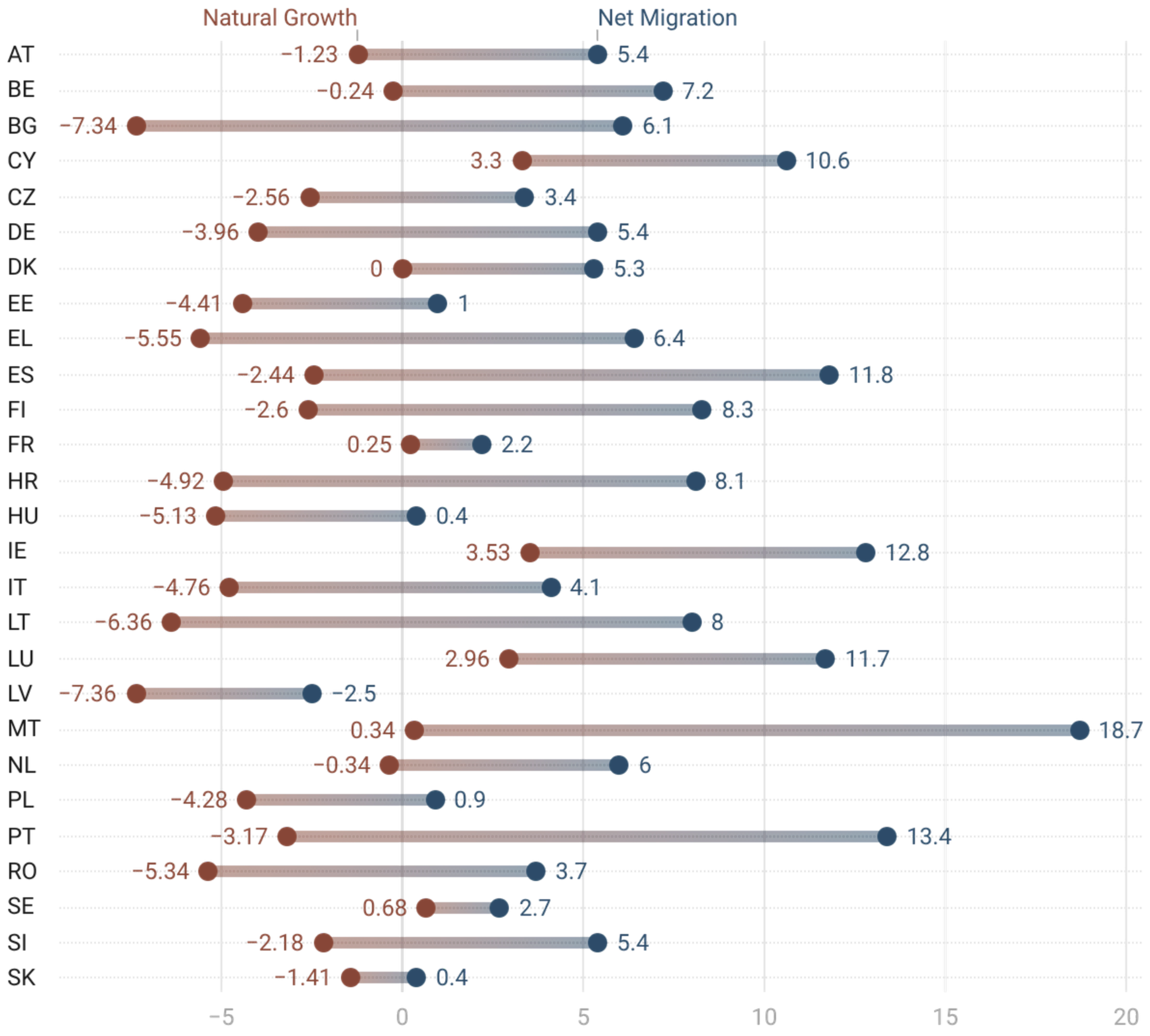
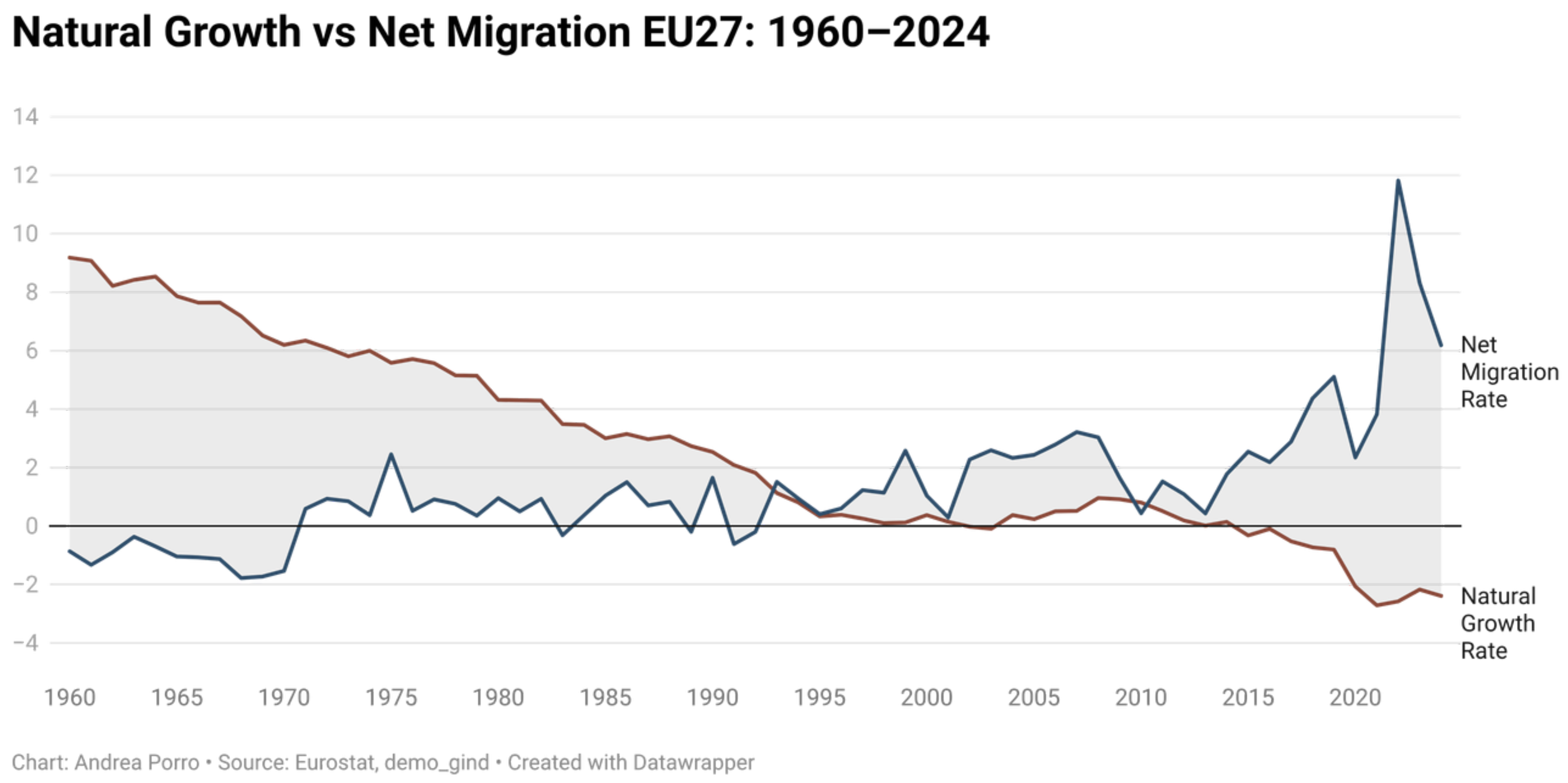


Chart: Andrea Porro • Source: Eurostat, demo_gind • Created with Datawrapper

The dot plot directly confronts the two forces shaping Europe's demographic balance. For each country, the **terracotta dot** represents the **natural growth rate**, the difference between births and deaths, while the **blue dot** represents the **net migration rate**. The wider the gap between the two dots, the greater the role migration plays in compensating for natural decline.

The pattern is striking: 20 out of 27 EU countries record a **negative natural growth rate**, meaning deaths outnumber births. The terracotta dots cluster heavily to the left of zero, while the blue dots sit consistently to the right, confirming that **migration is the primary** driver of population **growth** across the continent.

Can migration reverse the trend? From surplus to deficit



The chart shows the net migration rate across EU27 member states in 2024, the difference between people arriving and people leaving, expressed per 1.000 people.

In the 1960s, natural growth was the dominant force, with a rate of nearly +9 per 1,000 inhabitants, Europe's population was growing primarily through births. Net migration was negative as many Europeans were emigrating rather than arriving.

The two lines converge around 1995, when natural growth had fallen close to zero and migration began to emerge as the primary driver of population change. From this point onwards, the roles effectively reversed. Two extraordinary events are clearly visible in the most recent years. The COVID-19 pandemic caused a sharp spike in deaths in 2020, pushing natural growth to its most negative point on record. Just two years later, the war in Ukraine triggered an unprecedented surge in net migration in 2022 as millions of refugees entered EU territory.

Policy Implications

The findings of this analysis point to three interconnected policy priorities for Europe's demographic future.

1. **Reversing the fertility decline** requires sustained investment in family support policies such as: affordable childcare, parental leave, and housing access for young families. The modest recovery observed in the early 2000s suggests that structural incentives can have measurable effects, even if temporary.
2. **Managing the pension burden** will demand structural reforms across most EU27 countries. With dependency ratios projected to rise sharply and five countries already spending over 10% of GDP on pensions, a recalibration of retirement ages and pension system designs appears increasingly unavoidable.
3. **Harnessing migration strategically** is perhaps the most immediate lever available. With 20 out of 27 EU countries already dependent on migration to sustain their populations, coordinated EU-level migration and integration policies could help ensure that demographic benefits are distributed more evenly across member states.

Without coordinated action on these three fronts, Europe risks a future defined by shrinking workforces, strained public finances, and growing demographic disparities between member states.

Conclusion

Europe's aging population is not a future threat, it is a present reality. The data tell a consistent story across every indicator: a continent growing older, slower, and increasingly dependent on external population flows to sustain itself.

01.

How old is Europe really getting?

The data confirm that Europe is aging rapidly and continuously. Over the past six decades, the median age has risen across every single EU27 member state, without exception. However, the speed of this transition varies widely across the continent, revealing substantial demographic disparities with Italy leading with 49.1 years.

02.

What are the structural causes?

Two converging forces explain the aging trend. Fertility has remained below the replacement threshold of 2.1 since 1978, 47 consecutive years, reaching a new low of 1.45 in 2022. Simultaneously, life expectancy has risen to 80.9 years, creating a demographic scissors effect that continuously widens the gap between births and deaths.

03.

Can migration reverse the trend?

Migration has become Europe's primary demographic lifeline. 20 out of 27 EU countries now record negative natural growth, and migration is the only force preventing population decline in most member states. However, its sustainability is dependent on factors beyond demographic control.

LICENCE

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